Beamline breakdown. **The answer, spallation, relates both to beamlines and the act of breaking.**

My advisor told everyone recently: 'infiltrate accelerator lab synchrotrons'. Postdocs review overnight jets, examining cheap tickets. **The first letter of every word spells MATERIALS PROJECT.**

**Each acronym in '[ ]' represents a facility, and each acronym/phrase in '( )' represents a beamline/sector in that facility. Each line will clue a single number.**

(BioSAXS FlexX) [CHESS] **At CHESS, BioSAXS is beamline 7A1, and FlexX is beamline 7B2. Extract 77 from this.**
(IR-Micro-Spec Polymer-STXM HPISXS) [ALS] **At ALS, Infrared (IR) Micro-Spectroscopy is done on beamline 1.4, Polymer Scanning Transmission X-Ray Microscopy (STXM) is done on beamline 5.3.2.2, and High-Pressure In Situ Soft X-Ray Spectroscopy (HPISXS) is done on beamline 7.3.3. Extract 157 from this.**
(AMP XFM) [NSLS-II] **At NSLS-II, AMP will be on beamline 13-ID, and XFM is on beamline 4-BM. Extract 134 from this.**
(XSD-IMG-HST HEXM GSECARS) [APS] **At APS, XSD Imaging (IMG) with High-Speed Tomography (HST) is done on beamline 2-BM-A/B, HEXM is at beamline 20-ID, and GSECARS operates the beamlines at sector 13. Extract 22013 from this.**
(NSE LIQREF) [SNS] **At SNS in ORNL, NSE is beamline 15, and LIQREF is beamline 4. Extract 154 from this.**

**Thus, we have the numbers 77, 157, 134, 22013, and 154. On Materials Project, mp-77 is a S (Sulfur) material, mp-157 is a P (Phosphorous) material, mp-134 is an Al (Aluminum) material, mp-22013 is a $LaTiO_3$ (Lanthanum, Titanium, Oxygen) material, and mp-154 is a $N_2$ (Nitrogen) material. Putting this together gives SPAlLaTi$O_3N_2$, which we use to extract the final answer.**

Answer: **SPALLATION (10 letter word)**

## Related Readings

- https://en.wikipedia.org/wiki/Materials_Project
- https://next-gen.materialsproject.org/
- https://www.chess.cornell.edu/users/beamlines
- https://als.lbl.gov/beamlines/
- https://www.bnl.gov/nsls2/beamlines/
- https://www.aps.anl.gov/Beamlines/Beamline-Directory
- https://neutrons.ornl.gov/sns
- https://en.wikipedia.org/wiki/Spallation
- https://en.wikipedia.org/wiki/Spallation_Neutron_Source